

Second and Higher Derivatives: Motion, Decay, and Implicit Curves

Answer Key

AP Calculus AB/BC

Quick Answers

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| <p>1. $24t^2 - 30t$</p> <p>2. (a) $s''(t) = 6t - 18$, (b) $s''(5) = 12 \text{ m/s}^2$</p> <p>3. $48 \sin(2t)$</p> <p>4. $-4e^{\frac{(-1)t}{5}}$</p> <p>5. (a) $A''(t) = 12t^2 - 48t + 36$, (b) times with zero acceleration = 1, 3</p> | <p>6. (a) $dy/dx = \frac{1}{\sqrt{x}}$, (b) second derivative = $-\frac{1}{2x^{\frac{3}{2}}}$</p> <p>7. (a) second derivative = $-\frac{25}{(25-x^2)^{\frac{3}{2}}}$, (b) value at (3, 4) = $-\frac{25}{64}$</p> <p>8. (a) $P''(t) = 6t - 24$, (b) hour when P' is smallest = 4, —</p> |
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What is verified

13 of 14 answers machine-checked against SymPy, across 8 problems.
 — marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-3.A – problems 1, 2, 5, 8

FUN-3.C – problems 3, 4

FUN-3.D – problems 6, 7

Difficulty 1–5 of 5 across 14 tagged checks.

- 1.** A freight elevator's height is $h(t) = 2t^4 - 5t^3 + 7t$ metres, t in seconds. Find $h''(t)$.

Solution. Differentiate twice with the power rule.

$$h'(t) = 8t^3 - 15t^2 + 7$$

$$h''(t) = 24t^2 - 30t$$

The units of h'' are metres per second per second.

$h''(t) = 24t^2 - 30t$

- 2.** A test cart has position $s(t) = t^3 - 9t^2 + 24t$ metres, t in seconds. (a) Find $s''(t)$. (b) Find the acceleration at $t = 5$ seconds.

Solution (a). Velocity is the first derivative, acceleration the second.

$$s'(t) = 3t^2 - 18t + 24$$

$$s''(t) = 6t - 18$$

$$\boxed{s''(t) = 6t - 18}$$

Solution (b). Substitute $t = 5$ into the acceleration function: $s''(5) = 6(5) - 18 = 30 - 18$.

$$\boxed{s''(5) = 12 \text{ m/s}^2}$$

3. A buoy's height is $y(t) = 3 \sin(2t)$ metres, t in seconds. Find $y^{(4)}(t)$.

Solution. Each differentiation brings down a factor of 2 from the chain rule and advances the sine through its four-step cycle.

$$y'(t) = 6 \cos(2t)$$

$$y''(t) = -12 \sin(2t)$$

$$y'''(t) = -24 \cos(2t)$$

$$y^{(4)}(t) = 48 \sin(2t)$$

After four derivatives the sine returns, multiplied by $2^4 = 16$.

$$\boxed{y^{(4)}(t) = 48 \sin(2t)}$$

4. A cooling tank holds $V(t) = 500e^{-t/5}$ litres, t in minutes. Find $V'''(t)$.

Solution. Every derivative of $e^{-t/5}$ reproduces itself times $-\frac{1}{5}$, so three derivatives multiply the coefficient by $(-\frac{1}{5})^3 = -\frac{1}{125}$.

$$V'(t) = -100e^{-t/5}$$

$$V''(t) = 20e^{-t/5}$$

$$V'''(t) = -4e^{-t/5}$$

$$\boxed{V'''(t) = -4e^{-t/5}}$$

5. A drone's altitude is $A(t) = t^4 - 8t^3 + 18t^2 + 30$ metres, $0 \leq t \leq 6$ seconds. (a) Find $A''(t)$. (b) Find every time at which the vertical acceleration is zero.

Solution (a).

$$A'(t) = 4t^3 - 24t^2 + 36t$$

$$A''(t) = 12t^2 - 48t + 36$$

$$\boxed{A''(t) = 12t^2 - 48t + 36}$$

Solution (b). Set the acceleration to zero and factor: $12t^2 - 48t + 36 = 12(t^2 - 4t + 3) = 12(t - 1)(t - 3)$. Both roots lie in $[0, 6]$.

$$\boxed{t = 1 \text{ or } t = 3}$$

6. A cable is modelled by $y^2 = 4x$ with $y > 0$. (a) Find $\frac{dy}{dx}$ as a function of x . (b) Find $\frac{d^2y}{dx^2}$ as a function of x .

Solution (a). Differentiate both sides with respect to x , treating y as a function of x : $2y \frac{dy}{dx} = 4$, so $\frac{dy}{dx} = \frac{2}{y}$. On the branch $y > 0$ we have $y = 2\sqrt{x}$, hence

$$\boxed{\frac{dy}{dx} = \frac{2}{y} = \frac{1}{\sqrt{x}}}$$

Solution (b). Differentiate $\frac{dy}{dx} = x^{-1/2}$ once more:

$$\boxed{\frac{d^2y}{dx^2} = -\frac{1}{2x^{3/2}} = -\frac{4}{y^3}}$$

Either form is acceptable; the second is what implicit differentiation of $\frac{dy}{dx} = \frac{2}{y}$ produces directly.

7. The rim $x^2 + y^2 = 25$ (metres), upper half $y > 0$. (a) Find $\frac{d^2y}{dx^2}$ as a function of x . (b) Evaluate it at $(3, 4)$.

Solution (a). Implicitly, $2x + 2y \frac{dy}{dx} = 0$ gives $\frac{dy}{dx} = -\frac{x}{y}$. Differentiating that quotient and substituting $x^2 + y^2 = 25$ gives $-\frac{25}{y^3}$. On the upper half $y = \sqrt{25 - x^2}$, so

$$\boxed{\frac{d^2y}{dx^2} = -\frac{25}{(25 - x^2)^{3/2}} = -\frac{25}{y^3}}$$

Solution (b). At $(3, 4)$ the denominator is $y^3 = 4^3$, so the second derivative equals

$$\boxed{-\frac{25}{64}}$$

The negative value says the upper rim is concave down, which matches the picture.

8. A culture holds $P(t) = t^3 - 12t^2 + 36t + 50$ thousand cells after t hours, $0 \leq t \leq 8$. (a) Find $P''(t)$. (b) Find the hour at which the growth rate is smallest. (c) Explain how the sign of P'' settles it.

Solution (a).

$$P'(t) = 3t^2 - 24t + 36$$

$$P''(t) = 6t - 24$$

$$\boxed{P''(t) = 6t - 24}$$

Solution (b). The growth rate $P'(t)$ has its own critical point where its derivative $P''(t)$ is zero: $6t - 24 = 0$.

$$\boxed{t = 4}$$

That is 4 hours after the start, inside the interval.

Solution (c) — instructor-judged. A complete answer must argue from the sign of P'' , not from a table of values: $P''(t) = 6t - 24$ is negative for $t < 4$ and positive for $t > 4$, so the growth rate P' decreases and then increases, which makes $t = 4$ a minimum of P' . Award full credit only if the response is about the growth rate P' and not about the population P itself; P is increasing on both sides of $t = 4$, so a student who argues about P has answered a different question.